Announced Escalation with Strategic Drivers
A Two-Period Model with Latent Market Thickness

Abstract

We study a platform that commits to a first-period payment and a higher failure-contingent rescue payment. The platform observes market thickness but not realized supply. Riders decide whether to post and, after an initial failure, whether to abandon, repeat the initial payment, or activate the rescue payment. Drivers privately observe their costs and may reject the initial payment in anticipation of the rescue option. We first characterize the agents' anonymous symmetric pure-cutoff weak perfect Bayesian equilibrium correspondence under an arbitrary announced escalation policy. Each such equilibrium is summarized by a scalar driver cutoff: accepting immediately trades off current competition for assignment against the possibility of being paid more after all rival drivers also wait. We then embed this equilibrium response in the platform's completion-maximization problem. A flat payment admits a unique symmetric cutoff equilibrium. Starting from any active flat payment, a small announced escalation strictly increases completion whenever a positive mass of riders activates the rescue option. In the no-entry benchmark with positive incumbent survival, this local improvement holds at every positive market thickness; when $\beta > 1/2$, it strictly improves on the optimized flat payment. The value of escalation nevertheless vanishes in both very thin and very thick markets.

1 Introduction

Many platforms respond to an unfilled request by offering a higher payment in a later round. Announcing this escalation in advance improves the platform's ability to rescue a request, but it also changes first-round behavior: a driver may reject an acceptable payment today in the hope of receiving more after the request fails. The platform therefore cannot evaluate the second-round payment while holding first-round acceptance fixed.

We study this tradeoff in a two-period model. The platform observes a public measure of market thickness, denoted by m, but neither the platform policy nor the agents' strategies can condition on the realized number of available drivers. The platform commits to an announced escalation policy (p_1, p_2), with $p_2 \geq p_1$. A rider with private value decides whether to post the request. Drivers with private costs simultaneously decide whether to accept the first payment. If no driver accepts, the rider may abandon, repeat p_1, or activate p_2. Surviving incumbents and newly arriving drivers then make terminal acceptance decisions.

Our organization follows the policy–equilibrium–optimization logic used in the announced-pricing literature. In particular, Aviv and Pazgal (2008) first fix the seller's announced policy, solve the strategic customers' threshold response, and only then evaluate and optimize the seller's policy. We adopt the same order of analysis, while reversing the economic direction of the policy. Their consumers wait for a markdown and face a risk of stockout; our drivers wait for an escalation and face the risk that another driver accepts first. The common methodological point is that an announced future term must be evaluated through the equilibrium behavior it induces today.

The analysis proceeds in four steps. First, fixing an announced escalation policy, we solve the rider's continuation problem and the drivers' Bayesian acceptance game. The latter reduces to a scalar cutoff condition. Second, we insert that equilibrium response into the platform's completion probability and define the platform's policy problem. Third, we develop a flat-payment benchmark and compare it with a small announced escalation. Finally, we study how the value of the policy changes with market thickness and with fresh supply in the second period.

The main economic mechanism is a latent-competition wedge. A driver who accepts immediately competes with other accepting drivers for assignment. Her expected assignment share is

$$\phi(mp) = \frac{1 - e^{-mp}}{mp}.$$

A driver who waits reaches the rescue round only if every rival also rejects, an event with probability e^{-mp}. Hence

$$\phi(mp) - e^{-mp} > 0 \qquad (m > 0,\ p > 0).$$

When incumbents survive and a positive mass of riders activates rescue, this strict difference makes a marginal rescue payment valuable even when no new driver arrives in the second period.

2 Model

2.1 Operational setting and policy classes

There is one potential request and two decision periods. The public state is market thickness $m > 0$. Before any private information is observed, the platform commits to an announced failure-contingent rescue policy in

$$\mathcal{P}^D = \{(p_1, p_2) : 0 \le p_1 \le p_2 \le 1\}.$$

The payment p_j is paid by the rider to the selected driver. We treat this payment as a transfer and let the platform maximize the probability that a potential request is completed. A driver incurs cost only if she is selected; an accepting driver who is not selected receives zero and incurs no cost.

The policy is *announced*: after a first-period failure the platform cannot revise p_2. The second payment is also *optional*: the rider, rather than the platform, decides whether to activate it. A flat policy is the nested class

$$\mathcal{P}^F = \{(p, p) : 0 \le p \le 1\} \subset \mathcal{P}^D.$$

Thus the comparison is between two commitment protocols, not between an ex ante commitment and an ex post price revision.

2.2 Agents, latent supply, and information

From the rider's perspective, the number of incumbent drivers is

$$N^I \sim \mathrm{Pois}(m).$$

The realization of N^I is latent: it is not observed by the rider or by an individual driver, and the announced policy cannot depend on it.

We use the standard Poisson-population convention. Conditional on a focal driver being present, the number of *other* incumbent drivers remains $\mathrm{Pois}(m)$. This assumption is the finite-market implication of the Palm property of a Poisson population; it is not the ordinary conditional distribution $N^I - 1 \mid N^I \ge 1$.

The rider's value is $v \sim U[0,1]$. A completion in period 1 gives value v, whereas a completion in period 2 gives value βv, where $\beta \in (0,1)$. Each driver has an independently drawn cost $c \sim U[0,1]$. An incumbent remains available in period 2 with probability $\alpha \in [0,1]$. In addition, a fresh pool

$$N^E \sim \text{Pois}(\gamma m), \qquad \gamma \geq 0,$$

arrives in period 2, independently of all earlier uncertainty.

2.3 Sequence of events

The sequence is as follows.

1. The platform observes m and announces (p_1, p_2).
2. The rider observes v and decides whether to post.
3. The incumbent pool is realized. Each incumbent observes her own cost and simultaneously accepts or rejects p_1.
4. If at least one driver accepts, the platform selects one accepting driver uniformly and the request is completed.
5. If no driver accepts, the rider observes only the failure and chooses among abandoning, repeating p_1, and activating p_2.
6. Incumbent survival and fresh entry are realized. In the terminal period, all available drivers decide whether to accept the active payment. If at least one accepts, one is selected uniformly.

The rider does not observe the initial number of drivers, the number of active rejections, the survival realization, or the number of fresh arrivals.

Figure 1 gives the reduced extensive form. It suppresses the continuum of rider and driver types but preserves the order of moves, the public failure signal, and the information that remains latent at each stage.

2.4 Equilibrium and tie-breaking

We study weak perfect Bayesian equilibria (WPBE) in anonymous, symmetric, pure driver strategies. Strategies are sequentially rational, and beliefs after a first-period failure follow Bayes' rule. Because a failure has positive probability for every finite m, the relevant posterior is reached on path.

The following tie-breaking rules make the equilibrium assessment single valued at payoff ties. A rider with $v \geq p_1$ posts when indifferent, whereas a rider with $v < p_1$ does not post when indifferent. At the failure node, if all continuation actions yield zero, the rider repeats p_1 when $\beta v - p_1 \geq 0$ and abandons otherwise. When $p_2 = p_1$, repeat is selected over the nominal escalation action. We specify the action of a cutoff driver directly when stating the equilibrium conditions. The repeat–abandon convention is substantive: when $\gamma = 0$ and $a = p_1$, rider types with $v \geq p_1/\beta$ obtain zero from both actions, and their choice to repeat affects a deviating driver's continuation payoff.

3 Equilibrium Under an Announced Escalation Policy

We now fix m and an announced policy (p_1, p_2). Following the leader–follower order of the game, we first solve the agents' continuation behavior and then characterize first-period acceptance.

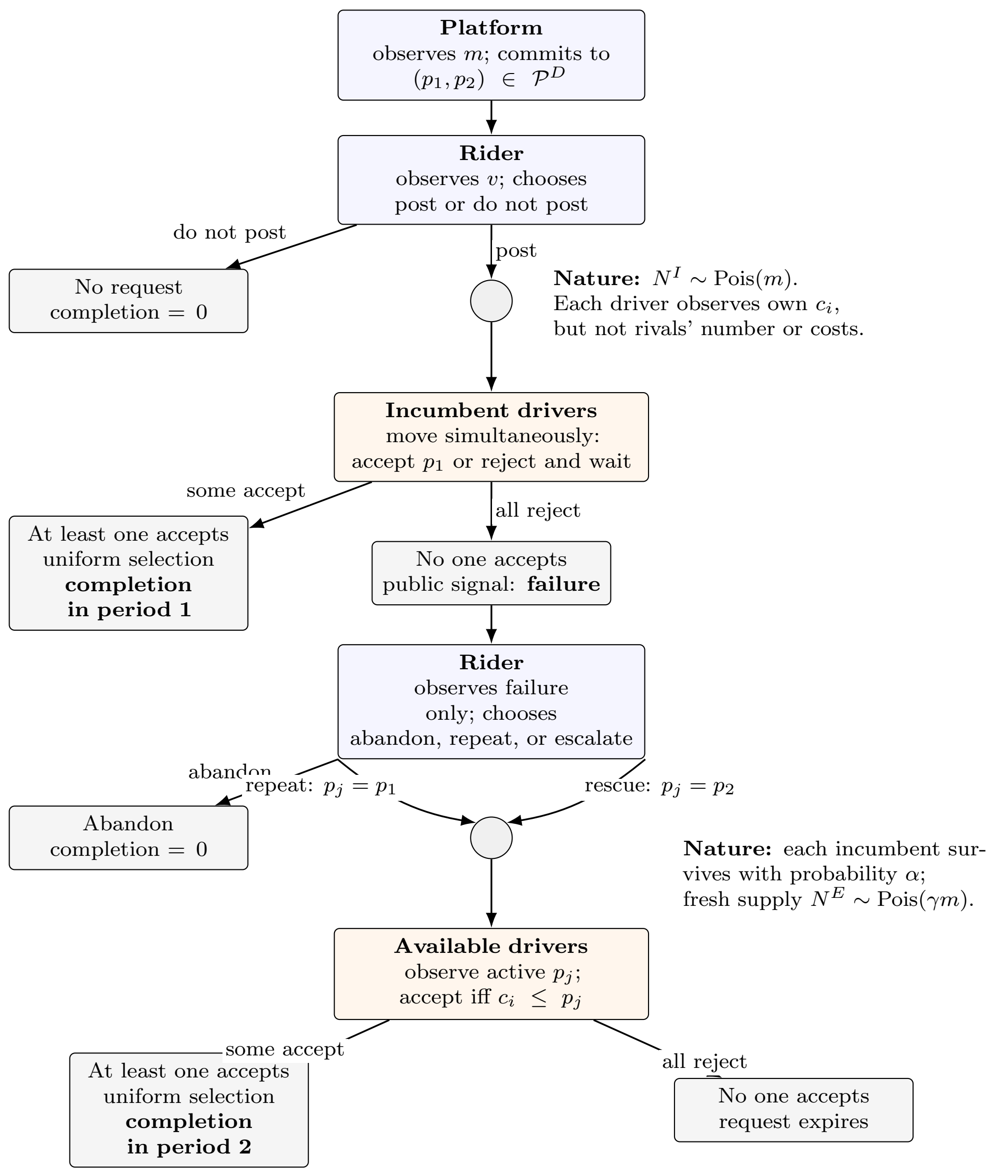

Figure 1: Reduced extensive form of the announced rescue game. Decision boxes identify the acting side; circles are chance moves. Continuous type branches and individual simultaneous driver actions are summarized inside their decision nodes. The public history consists of m, the announced policy, and whether period 1 failed; realized driver counts, rivals' costs, survival, and fresh entry remain latent.

3.1 Supply following a first-period failure

Define

$$\phi(x) = \begin{cases} (1-e^{-x})/x, & x>0, \\ 1, & x=0. \end{cases}$$

Suppose incumbents use a first-period cutoff $a \in [0, p_1]$: a driver with cost below a accepts, and a driver with cost above a rejects.

Lemma 1 (Poisson competition)**.** *Under cutoff a, the first-period completion probability conditional on posting is $1-e^{-ma}$. A focal driver who accepts in period 1 receives the assignment with expected share*

$$\sigma_m(a) = \phi(ma).$$

After a first-period failure, the number of drivers willing to accept payment p_j in period 2 is Poisson with mean

$$\lambda_j(m,a) = m\big[\alpha(p_j - a) + \gamma p_j\big].$$

Consequently, the period-2 coverage probability and a waiting incumbent's expected assignment share are

$$C_j(m,a) = 1 - e^{-\lambda_j(m,a)}, \qquad \widetilde{s}_{j,m}(a) = \phi(\lambda_j(m,a)).$$

The result follows from Poisson splitting and thinning. A failure reveals that no incumbent with cost below a is present. It does not change the distribution of incumbents with costs in $(a, p_j]$. Survival thins that group by α, while fresh supply contributes an independent Poisson group of mean $\gamma m p_j$.

3.2 The rider's equilibrium response

Given a, the expected payoff from posting is

$$\begin{aligned} \Pi(v;a) = {} & (1-e^{-ma})(v-p_1) \\ & + e^{-ma} \max\big\{0,\ C_1(m,a)(\beta v - p_1),\ C_2(m,a)(\beta v - p_2)\big\}. \end{aligned}$$

Under our tie-breaking convention, the rider posts if and only if

$$v \geq p_1.$$

Thus the posting mass is $1 - p_1$.

When $C_2 > C_1$, define the rider type indifferent between repeat and escalation by

$$v_m^M(a) = \frac{C_2(m,a)p_2 - C_1(m,a)p_1}{\beta[C_2(m,a) - C_1(m,a)]}.$$

When $C_2 = C_1$, set $v_m^M(a) = +\infty$. Since

$$v_m^M(a) \geq \frac{p_2}{\beta} \geq \frac{p_1}{\beta},$$

the continuation strategy takes a simple threshold form:

$$\begin{cases} \text{abandon}, & v < p_1/\beta, \\ \text{repeat } p_1, & p_1/\beta \leq v < v_m^M(a), \\ \text{activate } p_2, & v \geq v_m^M(a). \end{cases}$$

Conditional on posting, let

$$\eta_m^R(a) = \frac{[\min\{v_m^M(a), 1\} - p_1/\beta]^+}{1 - p_1}$$

and

$$\eta_m(a) = \frac{[1 - v_m^M(a)]^+}{1 - p_1}$$

denote the masses that repeat and escalate. Their sum is

$$\eta_m^R(a) + \eta_m(a) = \rho(p_1) := \frac{[1 - p_1/\beta]^+}{1 - p_1}.$$

The total continuation mass depends only on the first-period payment and rider patience. The cutoff, thickness, and rescue payment determine how this mass is divided between repeat and escalation.

3.3 The drivers' first-period game

A driver with $c \le p_1$ who accepts immediately obtains

$$U_m^A(c; a) = \phi(ma)(p_1 - c).$$

If she rejects, the request reaches period 2 only when all other first-period drivers also reject. Her waiting payoff is

$$\begin{aligned} U_m^W(c; a) = e^{-ma}\alpha\Big[&\eta_m^R(a)\phi(\lambda_1(m, a))[p_1 - c]^+ \\ &+ \eta_m(a)\phi(\lambda_2(m, a))[p_2 - c]^+\Big]. \end{aligned}$$

For $c \le p_1$, define the accept-minus-wait difference

$$\begin{aligned} \Psi_m(c; a) = \phi(ma)(p_1 - c) \\ - e^{-ma}\alpha\Big[&\eta_m^R(a)\phi(\lambda_1(m, a))(p_1 - c) \\ &+ \eta_m(a)\phi(\lambda_2(m, a))(p_2 - c)\Big]. \end{aligned}$$

The relevant single-crossing property is immediate:

$$\frac{\partial \Psi_m(c; a)}{\partial c} \le -\phi(ma) + e^{-ma}\alpha\rho(p_1) < 0$$

whenever $p_1 > 0$. Hence lower-cost drivers have a strictly stronger incentive to accept immediately. Drivers with $c > p_1$ reject because accepting gives a negative payoff while waiting gives a nonnegative payoff.

Let

$$f_m(a; p_1, p_2) := \Psi_m(a; a).$$

Proposition 1 (Symmetric cutoff WPBE)**.** *For every $m > 0$ and $0 \le p_1 \le p_2 \le 1$, with $p_1 < 1$, the announced policy admits an anonymous symmetric pure-strategy cutoff WPBE. The equilibrium cutoff belongs to*

$$\mathcal{E}_m^{\alpha,\gamma}(p_1, p_2) = \left\{ a \in [0, p_1] : \begin{array}{ll} f_m(0; p_1, p_2) \le 0, & a = 0, \\ f_m(a; p_1, p_2) = 0, & 0 < a < p_1, \\ f_m(p_1; p_1, p_2) \ge 0, & a = p_1 \end{array} \right\}.$$

This correspondence is nonempty and compact.

At $a = 0$, all types reject. At an interior cutoff, types below a accept and types above a reject. At $a = p_1$, all types with $c \le p_1$ accept. Existence follows from continuity and

$$f_m(p_1; p_1, p_2) = -e^{-mp_1} \alpha \eta_m(p_1) \phi(\lambda_2(m, p_1))(p_2 - p_1) \le 0.$$

If $f_m(0) \le 0$, reject-all is an equilibrium. If $f_m(0) > 0$ and $f_m(p_1) < 0$, continuity gives an interior root; if $f_m(p_1) = 0$, the boundary cutoff $a = p_1$ is an equilibrium.

4 The Platform's Policy Problem

Having solved the follower game for a fixed policy, we now evaluate the platform's objective. Given a cutoff equilibrium a, the unconditional completion probability is

$$\begin{aligned} M_m^{\alpha,\gamma}(p_1, p_2; a) = (1 - p_1) \Big[1 - e^{-ma} & \\ & + e^{-ma} \{ \eta_m^R(a) C_1(m, a) + \eta_m(a) C_2(m, a) \} \Big]. \end{aligned}$$

The expression separates the posting margin from execution:

$$\underbrace{1 - p_1}_{\text{posting}} \times \left[\underbrace{1 - e^{-ma}}_{\text{first-period completion}} + \underbrace{e^{-ma} (\eta_m^R C_1 + \eta_m C_2)}_{\text{failure-contingent rescue}} \right].$$

An arbitrary policy may admit more than one symmetric cutoff equilibrium. Accordingly, define its implementable completion set by

$$\mathcal{I}_m(p_1, p_2) = \left\{ M_m^{\alpha,\gamma}(p_1, p_2; a) : a \in \mathcal{E}_m^{\alpha,\gamma}(p_1, p_2) \right\}.$$

We use the conservative selection

$$\underline{M}_m^{\alpha,\gamma}(p_1, p_2) = \min_{a \in \mathcal{E}_m^{\alpha,\gamma}(p_1, p_2)} M_m^{\alpha,\gamma}(p_1, p_2; a)$$

and define the announced-escalation value

$$D^*_{\alpha,\gamma}(m) = \sup_{0 \le p_1 \le p_2 \le 1} \underline{M}_m^{\alpha,\gamma}(p_1, p_2).$$

At $p_1 = 1$, completion is defined to be zero. The minimum above is only over anonymous symmetric pure cutoff equilibria under the stated rider tie-breaking. It should therefore be interpreted as a worst *symmetric-cutoff* equilibrium criterion, not as a minimum over all possible mixed or asymmetric WPBE.

5 A Flat-Payment Benchmark

We next develop the benchmark against which announced escalation is evaluated. Set $p_1 = p_2 = p$.

Proposition 2 (Unique flat equilibrium)**.** *For every $m > 0$ and $p > 0$, a flat policy admits a unique symmetric cutoff WPBE:*

$$\mathcal{E}_m^{\alpha,\gamma}(p, p) = \{p\}.$$

To see why, for every $a < p$,

$$f_m(a; p, p) = (p - a)\left[\phi(ma) - e^{-ma}\alpha\rho(p)\phi\big(m[\alpha(p - a) + \gamma p]\big)\right] > 0.$$

Thus no cutoff below p can be sustained. A driver who rejects p has cost above p, so surviving incumbents do not accept the same payment in period 2. Any second-period completion under a flat policy comes from fresh drivers.

The flat completion probability is

$$Q^F_\gamma(m, p) = (1 - p)(1 - e^{-mp}) + e^{-mp}[1 - p/\beta]^+(1 - e^{-\gamma mp}).$$

The optimized flat value is

$$F^*_\gamma(m) = \sup_{p \in [0,1]} Q^F_\gamma(m, p).$$

Because flat policies are feasible announced policies,

$$D^*_{\alpha,\gamma}(m) \geq F^*_\gamma(m).$$

6 The Value of Announced Escalation

We now compare a flat policy with a nearby announced escalation. Fix

$$p_1 = p, \qquad p_2 = p + \varepsilon, \qquad p \in (0, \beta),$$

and let $\varepsilon \downarrow 0$. Define

$$u = 1 - p, \qquad R = e^{-mp}, \qquad E = e^{-\gamma mp},$$
$$\sigma = \phi(mp), \qquad \ell = \phi(\gamma mp), \qquad \rho = \frac{1 - p/\beta}{1 - p}.$$

When $\alpha + \gamma > 0$, define the limiting rider threshold

$$\bar{v}_m = \frac{1}{\beta}\left[p + \frac{1 - E}{mE(\alpha + \gamma)}\right]$$

and the limiting escalation mass

$$\eta^0_m = \frac{[1 - \bar{v}_m]^+}{1 - p}.$$

Theorem 1 (Local equilibrium response)**.** *Suppose $m > 0$, $p \in (0, \beta)$, $\alpha + \gamma > 0$, and $\bar{v}_m \neq 1$. For all sufficiently small $\varepsilon > 0$, the policy $(p, p + \varepsilon)$ admits a unique symmetric cutoff WPBE. Its cutoff satisfies*

$$a_\varepsilon = p - \kappa_m \varepsilon + o(\varepsilon),$$

where

$$\kappa_m = \frac{R\alpha\ell\,\eta^0_m}{\sigma - R\alpha\ell\rho}.$$

Moreover, every symmetric cutoff equilibrium satisfies

$$0 \leq p - a \leq \frac{\alpha\rho}{1 - \alpha\rho}\varepsilon.$$

The localization bound is important. It rules out an equilibrium branch far from the flat cutoff and makes the local comparison robust to symmetric-cutoff equilibrium selection. The proof first rescales the equilibrium condition using $k = (p - a)/\varepsilon$; a direct application of the implicit-function theorem at the flat policy is invalid because the rider's escalation threshold has a $0/0$ limit there.

Define

$$B_m = 1 - \rho[1 - (1 - \alpha)E].$$

Theorem 2 (An active rescue payment raises completion)**.** *Under the assumptions of Theorem 1,*

$$\begin{aligned} L_m^{\alpha,\gamma}(p) &:= \lim_{\varepsilon \downarrow 0} \frac{M_m^{\alpha,\gamma}(p, p+\varepsilon) - Q_\gamma^F(m,p)}{\varepsilon} \\ &= m(1-p)R\left[E(\alpha+\gamma)\eta_m^0 - \kappa_m B_m\right]. \end{aligned}$$

If $\bar{v}_m < 1$, then

$$L_m^{\alpha,\gamma}(p) > 0.$$

If $\bar{v}_m > 1$, then $\eta_m^0 = \kappa_m = 0$ and

$$L_m^{\alpha,\gamma}(p) = 0.$$

The coefficient has a direct decomposition. The term

$$m(1-p)RE(\alpha+\gamma)\eta_m^0$$

is the marginal rescue gain generated by greater period-2 coverage. The term

$$m(1-p)R\kappa_m B_m$$

is the strategic-delay loss caused by a lower first-period driver cutoff. The theorem shows that whenever a positive mass of riders uses the marginal rescue option, the coverage gain strictly dominates the induced delay.

6.1 No fresh entry

The cleanest benchmark sets $\gamma = 0$. Then

$$E = \ell = 1, \qquad \bar{v}_m = p/\beta < 1, \qquad \eta_m^0 = \rho.$$

The local coefficient simplifies to

$$L_m^{\alpha,0}(p) = \frac{m(1-p)e^{-mp}\alpha\rho[\phi(mp) - e^{-mp}]}{\phi(mp) - e^{-mp}\alpha\rho} > 0.$$

The strict inequality is governed by

$$\phi(mp) - e^{-mp} > 0.$$

The first term is the assignment share from accepting now; the second is the probability that all rivals reject and thereby allow a waiting driver to reach period 2.

Corollary 1 (Strict improvement over the optimized flat policy)**.** *Suppose $\gamma = 0$, $\alpha > 0$, and $\beta > 1/2$. Then*

$$D_{\alpha,0}^*(m) > F_0^*(m) \qquad \textit{for every } m > 0.$$

When $\gamma = 0$,

$$Q_0^F(m,p) = (1-p)(1-e^{-mp}).$$

Its unique optimizer $p_F(m)$ satisfies

$$e^{mp_F(m)} = 1 + m[1 - p_F(m)]$$

and

$$0 < p_F(m) < \frac{1}{2}.$$

Thus $p_F(m) < \beta$, and Theorem 2 can be applied directly at the optimized flat payment. The closed form

$$p_F(m) = \frac{m + 1 - W_0(e^{m+1})}{m}$$

is useful for computation but not needed for the economic argument.

7 Market Thickness

Define the value of announced escalation by

$$V_{\alpha,\gamma}(m) = D^*_{\alpha,\gamma}(m) - F^*_\gamma(m).$$

We first establish the two endpoint results.

Proposition 3 (Thickness endpoints)**.** *For every $\alpha \in [0,1]$, $\gamma \geq 0$, and $\beta \in (0,1)$,*

$$\lim_{m \downarrow 0} D^*_{\alpha,\gamma}(m) = \lim_{m \downarrow 0} F^*_\gamma(m) = \lim_{m \downarrow 0} V_{\alpha,\gamma}(m) = 0,$$

and

$$\lim_{m \to \infty} D^*_{\alpha,\gamma}(m) = \lim_{m \to \infty} F^*_\gamma(m) = 1, \qquad \lim_{m \to \infty} V_{\alpha,\gamma}(m) = 0.$$

For the thin-market limit, any completion requires at least one driver in the union of the initial and fresh pools, so

$$D^*_{\alpha,\gamma}(m) \leq 1 - e^{-(1+\gamma)m}.$$

For the thick-market limit, the flat payment $p = m^{-1/2}$ yields first-period completion at least

$$(1 - m^{-1/2})(1 - e^{-\sqrt{m}}),$$

which converges to one.

Under $\gamma = 0$, $\alpha > 0$, and $\beta > 1/2$, combining Proposition 3 with Corollary 1 gives the current thickness result:

$$V_{\alpha,0}(m) > 0 \quad \text{for every finite } m > 0,$$

but

$$V_{\alpha,0}(m) \to 0 \quad \text{as } m \downarrow 0 \text{ or } m \to \infty.$$

This is an intermediate-thickness result in the endpoint sense. It does not by itself establish continuity, existence of an attained interior maximizer, strict single-peakedness, or uniqueness of the maximizing thickness.

8 Fresh Supply and the Order of the Gain

Fresh entry changes the thin-market order of the local gain. With no fresh entry,

$$L_m^{\alpha,0}(p) = \frac{\alpha\rho\, p(1-p)}{2(1-\alpha\rho)} m^2 + O(m^3).$$

The value is second order because a thin market must contain an incumbent and that incumbent must perceive a positive probability of a rival.

For $\gamma > 0$, define

$$\bar{v}_0 = \frac{p(\alpha + 2\gamma)}{\beta(\alpha + \gamma)}$$

and

$$\eta_0^0 = \frac{[1 - \bar{v}_0]^+}{1 - p}.$$

If

$$p < \frac{\beta(\alpha + \gamma)}{\alpha + 2\gamma},$$

then $\eta_0^0 > 0$, and

$$L_m^{\alpha,\gamma}(p) = m(1-p)\gamma\eta_0^0 + O(m^2).$$

Fresh supply therefore raises the local gain from order m^2 to order m whenever the rescue option remains active in the thin-market limit. These expansions concern the local coefficient at a fixed flat payment; they do not yet characterize the asymptotic order of the globally optimized value $V_{\alpha,\gamma}(m)$.

9 Discussion

The model can be read through three nested environments. With $\alpha = 1$ and $\gamma = 0$, the driver pool is closed and all incumbents remain available. With $0 < \alpha < 1$ and $\gamma = 0$, exit weakens rescue capacity but does not eliminate the strict value of escalation. With $\gamma > 0$, replenishment creates an additional rescue channel and may generate a first-order thin-market gain.

Several issues are deliberately left outside the current theorem set. We do not claim global uniqueness for every dynamic policy, characterize mixed or asymmetric WPBE, or prove that the platform's global supremum is attained. Likewise, the endpoint result does not establish a unique or strictly single-peaked value profile. These questions concern global equilibrium selection and the platform's design correspondence rather than the local escalation mechanism established here.

The economic comparison with announced markdown models is useful but not literal. A forward-looking buyer delays purchase to obtain a lower price and risks losing availability. Here a strategic supplier delays acceptance to obtain a higher payment and risks losing the request to a rival. This change in direction turns the availability risk into a competition-for-assignment wedge and makes latent market thickness the central state variable.

A Proof Roadmap and Technical Details

The main text is organized around the platform's policy problem. For reference, the proof architecture is as follows; keeping these technical steps in the appendix preserves the policy–equilibrium–comparison flow of the main text.

A.1 Poisson splitting, thinning, and the assignment-share identity in Lemma 1.

A.2 Rider posting and continuation thresholds.

A.3 Driver single crossing, the boundary cutoff conditions, and existence in Proposition 1.

A.4 Uniqueness of the flat cutoff in Proposition 2.

A.5 Localization for $(p, p+\varepsilon)$, followed by the rescaled equation $k = (p-a)/\varepsilon$, proving Theorem 1.

A.6 Differentiation of completion along the localized equilibrium branch and the positivity inequality in Theorem 2.

A.7 The optimal flat-payment first-order condition and Corollary 1.

A.8 The endpoint bounds in Proposition 3.

References

Aviv, Y., and A. Pazgal. 2008. Optimal pricing of seasonal products in the presence of forward-looking consumers. *Manufacturing & Service Operations Management* 10(3), 339–359. https://doi.org/10.1287/msom.1070.0183.